Phase-adjusted realification of a $\mathbb{C}^3$ Kochen-Specker configuration into $\mathbb{R}^6$

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We describe a general method that embeds any finite set of rays in $\mathbb{C}^3$ into $\mathbb{R}^6$ as real vectors in such a way that all complex orthogonality relations are preserved, and no new real orthogonalities are introduced, by exploiting the phase freedom of each ray. This yields a faithful orthogonal representation of the original complex configuration. As a concrete input, we consider the 165 projectively distinct rays used in a $\mathbb{C}^3$ Kochen–Specker configuration obtained from mutually unbiased bases. We list these 165 rays explicitly in $\mathbb{C}^3$ and give, for each of them, its explicit image in $\mathbb{R}^6$ under the standard coordinate-wise realification map.

I. A KOCHEN-SPECKER PROOF FOR THE NONEQUIVALENCE OF REAL AND COMPLEX HILBERT SPACE

A foundational question in quantum theory is whether the use of complex numbers is a matter of mathematical convenience or a fundamental necessity. The standard approach often treats real and complex Hilbert spaces as largely equivalent, particularly through a procedure known as “realification,” where any system in a complex Hilbert space $\mathbb{C}^n$ can be represented in a real space of doubled dimensionality, $\mathbb{R}^{2n}$. This perspective has been challenged from multiple angles, suggesting that the structure of complex Hilbert spaces permits physical phenomena and logical relationships that have no counterpart in the real domain.

Suggestions for exploring this inequivalence by mapping complex quantum configurations into higher-dimensional real spaces have been proposed [1] and have found compelling statistical corroboration. For instance, experiments have shown that correlations generated by quantum systems described in complex spaces can violate constraints derived for any theory based on real Hilbert spaces [2], a topic that has sparked broader discussion on the physical role of imaginary numbers [3].

While statistical arguments provide powerful evidence, this paper presents a deterministic proof based on the logical framework of the Kochen-Specker (KS) theorem. Rather than relying on inequalities and probabilities, the KS approach demonstrates impossibility through a direct contradiction arising from the geometric arrangement of vectors and their orthogonality relations. We employ a specific KS-type configuration developed by Adán Cabello [4], which was recently analyzed by John Harding and Remi Salinas Schmeis [5].

The key to this proof lies in a structural feature that starkly separates three-dimensional complex and real Hilbert spaces: the existence of *Mutually Unbiased Bases* (MUBs). Two orthonormal bases are mutually unbiased if a measurement in one basis completely randomizes the outcomes of a measurement prepared in any state of the other basis. The theory of MUBs is a rich field with deep connections to quantum information and foundational mathematics [6–11]. The crucial facts for this argument are:

1. In the complex Hilbert space $\mathbb{C}^3$, a complete set of $D+1=4$ MUBs can be constructed.
2. In the real Hilbert space $\mathbb{R}^3$, it is mathematically impossible to find even two MUBs.

The KS configuration we analyze is built directly from the four MUBs available in $\mathbb{C}^3$. The intricate network of orthogonality relations among the vectors in this set is a direct consequence of the unique geometric structure afforded by these bases. This specific set of complex quantum observables—represented by projectors onto the vectors of the configuration—therefore constitutes a valid proof of quantum contextuality.

The final step of the argument is to recognize that this entire construction cannot be reproduced in any real Hilbert space, regardless of its dimension [12]. The foundational elements—the four MUBs in three dimensions—simply do not exist in the real domain. Consequently, the logical structure of the resulting KS set is intrinsically complex. It serves as a concrete, deterministic example of a set of physical propositions and their relationships that can be formulated in complex quantum mechanics but is fundamentally forbidden in a theory restricted to real numbers. This provides a direct proof that, at the level of its logical structure, three-dimensional complex quantum theory is fundamentally richer and inequivalent to its real-valued counterpart.

II. PHASE-ADJUSTED REALIFICATION IN $\mathbb{C}^3$

A. Set-up

Let $\{\,[\psi_k]\,\}_{k=1}^{N}$ be a finite set of rays in $\mathbb{C}^3$, with normalized representatives $\psi_k \in \mathbb{C}^3$, and let

$$c_{k\ell} := \langle \psi_k, \psi_\ell \rangle_{\mathbb{C}} \tag{1}$$

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denote their complex inner products.

For each ray we may multiply ψ_k by an arbitrary unit complex phase $e^{\mathrm{i}\theta_k}$ without changing the ray. We exploit this freedom and define a *phase-adjusted realification*

$$R_k := \Phi_0\big(e^{\mathrm{i}\theta_k}\,\psi_k\big) \in \mathbb{R}^6, \tag{2}$$

where

$$\Phi_0 : \mathbb{C}^3 \to \mathbb{R}^6, \quad (z_1, z_2, z_3) \mapsto (\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3). \tag{3}$$

Then

$$\begin{aligned} R_k \cdot R_\ell &= \Phi_0(e^{\mathrm{i}\theta_k}\psi_k) \cdot \Phi_0(e^{\mathrm{i}\theta_\ell}\psi_\ell) \\ &= \Re\Big\langle e^{\mathrm{i}\theta_k}\psi_k, e^{\mathrm{i}\theta_\ell}\psi_\ell\Big\rangle_{\mathbb{C}} \\ &= \Re\big(e^{\mathrm{i}(\theta_\ell-\theta_k)}c_{k\ell}\big). \end{aligned} \tag{4}$$

If $c_{k\ell} = 0$, then $R_k \cdot R_\ell = 0$ for all phases, so complex orthogonality is always preserved. The only potential issue is that for $c_{k\ell} \neq 0$ we might accidentally choose phases such that $R_k \cdot R_\ell = 0$, thereby introducing a spurious orthogonality.

B. Forbidden phase differences for non-orthogonal pairs

Assume $c_{k\ell} \neq 0$. Write $c_{k\ell} = |c_{k\ell}|e^{\mathrm{i}\varphi_{k\ell}}$. Then Eq. (4) becomes

$$R_k \cdot R_\ell = |c_{k\ell}|\,\cos\big((\theta_\ell - \theta_k) + \varphi_{k\ell}\big). \tag{5}$$

Thus $R_k \cdot R_\ell = 0$ if and only if

$$(\theta_\ell - \theta_k) + \varphi_{k\ell} \equiv \frac{\pi}{2} \pmod{\pi}. \tag{6}$$

For a fixed θ_k and fixed nonzero $c_{k\ell}$, this excludes at most two values of θ_ℓ modulo 2π:

$$\theta_\ell \equiv -\varphi_{k\ell} + \frac{\pi}{2} \quad \text{or} \quad \theta_\ell \equiv -\varphi_{k\ell} + \frac{3\pi}{2} \pmod{2\pi}. \tag{7}$$

C. Existence of a faithful embedding

We construct the phases $\theta_1, \ldots, \theta_N$ inductively.

- Set $\theta_1 := 0$ arbitrarily.
- Suppose $\theta_1, \ldots, \theta_{m-1}$ have been chosen such that for all non-orthogonal pairs (k, ℓ) with $1 \le k < \ell < m$ we have $R_k \cdot R_\ell \neq 0$.
- For the new ray m, consider all pairs (k, m) with $1 \le k < m$ and $c_{km} \neq 0$. For each such pair, the two forbidden values of θ_m described above define a finite set $F_{k,m} \subset S^1$.

Let

$$F_m := \bigcup_{\substack{1\le k<m \\ c_{km}\neq 0}} F_{k,m}, \tag{8}$$

which is a finite subset of the unit circle. Choose θ_m anywhere on $S^1 \setminus F_m$.

By construction, for every non-orthogonal pair (k, m) we have $R_k \cdot R_m \neq 0$, while pairs with $c_{km} = 0$ remain orthogonal for all phases. Induction in m shows that we obtain a full set of phases $\{\theta_k\}_{k=1}^N$ such that

$$c_{k\ell} = 0 \iff R_k \cdot R_\ell = 0. \tag{9}$$

Thus $\{R_k\}_{k=1}^N \subset \mathbb{R}^6$ is a faithful orthogonal representation of the original finite set of rays in $\mathbb{C}^3$.

In particular, if $\{\psi_i, \psi_j, \psi_k\}$ form an orthonormal basis of $\mathbb{C}^3$, then each pair is orthogonal in $\mathbb{C}^3$, hence each pair R_i, R_j, R_k is orthogonal in $\mathbb{R}^6$, so orthogonal triples (contexts) are preserved.

III. EXPLICIT LIST OF THE 165 RAYS AND THEIR IMAGES IN $\mathbb{R}^6$

A. Complex notation

We use the primitive third root of unity

$$\omega := e^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \mathrm{i}\,\frac{\sqrt{3}}{2}, \qquad \omega^2 = \overline{\omega} = -\frac{1}{2} - \mathrm{i}\,\frac{\sqrt{3}}{2}, \tag{10}$$

with $1 + \omega + \omega^2 = 0$.

Every coordinate in the 165 rays below is a real multiple of 1, ω, or ω^2 with integer coefficients in $\{-2, -1, 0, 1, 2\}$. For any $z = a + b\,\omega + c\,\omega^2$ $(a, b, c \in \mathbb{R})$ we have

$$\Re z = a - \frac{b+c}{2}, \tag{11}$$

$$\Im z = \frac{\sqrt{3}}{2}\,(b - c). \tag{12}$$

In particular, $0 \mapsto (0, 0)$, and

$$\begin{aligned} 1 &\mapsto (1, 0), & -1 &\mapsto (-1, 0), \\ \omega &\mapsto \Big(-\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2}\Big), & \omega^2 &\mapsto \Big(-\tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2}\Big), \\ 2\omega &\mapsto \big(-1,\ \sqrt{3}\big), & 2\omega^2 &\mapsto \big(-1,\ -\sqrt{3}\big), \\ -\omega &\mapsto \Big(\tfrac{1}{2},\ -\tfrac{\sqrt{3}}{2}\Big), & -\omega^2 &\mapsto \Big(\tfrac{1}{2},\ \tfrac{\sqrt{3}}{2}\Big), \\ -2\omega &\mapsto \big(1,\ -\sqrt{3}\big), & -2\omega^2 &\mapsto \big(1,\ \sqrt{3}\big). \end{aligned} \tag{13}$$

B. Explicit realification

For concreteness, we list the real images of all 165 rays. Each row of Table I contains a label, the corresponding ray in $\mathbb{C}^3$, and its image in $\mathbb{R}^6$ under the canonical realification map (3) (with no additional phase factors). For a vector $(z_1, z_2, z_3) \in \mathbb{C}^3$ we write its real image as

$$(\Re z_1, \Re z_2, \Re z_3, \Im z_1, \Im z_2, \Im z_3) \in \mathbb{R}^6. \tag{14}$$

TABLE I: All 165 rays in $\mathbb{C}^3$ and their images in $\mathbb{R}^6$ under the canonical realification (3).

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
a_{11}	$(1,0,0)$	$(1,0,0,\ 0,0,0)$
a_{21}	$(0,1,0)$	$(0,1,0,\ 0,0,0)$
a_{31}	$(0,0,1)$	$(0,0,1,\ 0,0,0)$
u_1	$(1,1,1)$	$(1,1,1,\ 0,0,0)$
u_2	$(1,\omega,\omega^2)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
u_3	$(1,\omega^2,\omega)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
u_4	$(1,\omega,\omega)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
u_5	$(1,\omega^2,1)$	$\left(1,-\frac{1}{2},1,\ 0,-\frac{\sqrt{3}}{2},0\right)$
u_6	$(1,1,\omega^2)$	$\left(1,1,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
u_7	$(1,\omega^2,\omega^2)$	$\left(1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
u_8	$(1,\omega,1)$	$\left(1,-\frac{1}{2},1,\ 0,\frac{\sqrt{3}}{2},0\right)$
u_9	$(1,1,\omega)$	$\left(1,1,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
b_{11}	$(0,1,1)$	$(0,1,1,\ 0,0,0)$
b_{12}	$(0,\omega,\omega^2)$	$\left(0,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{13}	$(0,\omega^2,\omega)$	$\left(0,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{21}	$(1,0,1)$	$(1,0,1,\ 0,0,0)$
b_{22}	$(1,0,\omega^2)$	$\left(1,0,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
b_{23}	$(1,0,\omega)$	$\left(1,0,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
b_{31}	$(1,1,0)$	$(1,1,0,\ 0,0,0)$
b_{32}	$(1,\omega,0)$	$\left(1,-\frac{1}{2},0,\ 0,\frac{\sqrt{3}}{2},0\right)$
b_{33}	$(1,\omega^2,0)$	$\left(1,-\frac{1}{2},0,\ 0,-\frac{\sqrt{3}}{2},0\right)$
c_{11}	$(0,1,-1)$	$(0,1,-1,\ 0,0,0)$
c_{12}	$(0,\omega,-\omega^2)$	$\left(0,-\frac{1}{2},\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
c_{13}	$(0,\omega^2,-\omega)$	$\left(0,-\frac{1}{2},\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
c_{21}	$(-1,0,1)$	$(-1,0,1,\ 0,0,0)$
c_{22}	$(-\omega,0,1)$	$\left(\frac{1}{2},0,1,\ -\frac{\sqrt{3}}{2},0,0\right)$
c_{23}	$(-\omega^2,0,1)$	$\left(\frac{1}{2},0,1,\ \frac{\sqrt{3}}{2},0,0\right)$
c_{31}	$(1,-1,0)$	$(1,-1,0,\ 0,0,0)$
c_{32}	$(\omega^2,-1,0)$	$\left(-\frac{1}{2},-1,0,\ -\frac{\sqrt{3}}{2},0,0\right)$
c_{33}	$(\omega,-1,0)$	$\left(-\frac{1}{2},-1,0,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{11}	$(-2,1,1)$	$(-2,1,1,\ 0,0,0)$
d_{12}	$(-2,\omega,\omega^2)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
d_{13}	$(-2,\omega^2,\omega)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
d_{14}	$(-2,\omega,\omega)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
d_{15}	$(-2,\omega^2,1)$	$\left(-2,-\frac{1}{2},1,\ 0,-\frac{\sqrt{3}}{2},0\right)$
d_{16}	$(-2,1,\omega^2)$	$\left(-2,1,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
d_{17}	$(-2,\omega^2,\omega^2)$	$\left(-2,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
d_{18}	$(-2,\omega,1)$	$\left(-2,-\frac{1}{2},1,\ 0,\frac{\sqrt{3}}{2},0\right)$
d_{19}	$(-2,1,\omega)$	$\left(-2,1,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
d_{21}	$(1,-2,1)$	$(1,-2,1,\ 0,0,0)$
d_{22}	$(\omega^2,-2,\omega)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
d_{23}	$(\omega,-2,\omega^2)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
d_{24}	$(\omega^2,-2,1)$	$\left(-\frac{1}{2},-2,1,\ -\frac{\sqrt{3}}{2},0,0\right)$
d_{25}	$(\omega,-2,\omega)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
d_{26}	$(1,-2,\omega^2)$	$\left(1,-2,-\frac{1}{2},\ 0,0,-\frac{\sqrt{3}}{2}\right)$
d_{27}	$(\omega,-2,1)$	$\left(-\frac{1}{2},-2,1,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{28}	$(\omega^2,-2,\omega^2)$	$\left(-\frac{1}{2},-2,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
d_{29}	$(1,-2,\omega)$	$\left(1,-2,-\frac{1}{2},\ 0,0,\frac{\sqrt{3}}{2}\right)$
d_{31}	$(1,1,-2)$	$(1,1,-2,\ 0,0,0)$
d_{32}	$(\omega,\omega^2,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
d_{33}	$(\omega^2,\omega,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
d_{34}	$(\omega^2,1,-2)$	$\left(-\frac{1}{2},1,-2,\ -\frac{\sqrt{3}}{2},0,0\right)$
d_{35}	$(1,\omega^2,-2)$	$\left(1,-\frac{1}{2},-2,\ 0,-\frac{\sqrt{3}}{2},0\right)$
d_{36}	$(\omega,\omega,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
d_{37}	$(\omega,1,-2)$	$\left(-\frac{1}{2},1,-2,\ \frac{\sqrt{3}}{2},0,0\right)$
d_{38}	$(1,\omega,-2)$	$\left(1,-\frac{1}{2},-2,\ 0,\frac{\sqrt{3}}{2},0\right)$
d_{39}	$(\omega^2,\omega^2,-2)$	$\left(-\frac{1}{2},-\frac{1}{2},-2,\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{121}	$(1,1,-1)$	$(1,1,-1,\ 0,0,0)$
b_{122}	$(1,\omega,-\omega^2)$	$\left(1,-\frac{1}{2},\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{123}	$(1,\omega^2,-\omega)$	$\left(1,-\frac{1}{2},\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{124}	$(\omega,\omega^2,-\omega^2)$	$\left(-\frac{1}{2},-\frac{1}{2},\frac{1}{2},\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{125}	$(\omega,1,-\omega)$	$\left(-\frac{1}{2},1,\frac{1}{2},\ \frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
b_{126}	$(\omega,\omega,-1)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
b_{127}	$(\omega^2,\omega,-\omega)$	$\left(-\frac{1}{2},-\frac{1}{2},\frac{1}{2},\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{128}	$(\omega^2,1,-\omega^2)$	$\left(-\frac{1}{2},1,\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
b_{129}	$(\omega^2,\omega^2,-1)$	$\left(-\frac{1}{2},-\frac{1}{2},-1,\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{131}	$(1,-1,1)$	$(1,-1,1,\ 0,0,0)$
b_{132}	$(1,-\omega,\omega^2)$	$\left(1,\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{133}	$(1,-\omega^2,\omega)$	$\left(1,\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{134}	$(\omega,-\omega^2,\omega^2)$	$\left(-\frac{1}{2},\frac{1}{2},-\frac{1}{2},\ \frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{135}	$(\omega,-1,\omega)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ \frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$
b_{136}	$(\omega,-\omega,1)$	$\left(-\frac{1}{2},\frac{1}{2},1,\ \frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},0\right)$
b_{137}	$(\omega^2,-\omega,\omega)$	$\left(-\frac{1}{2},\frac{1}{2},-\frac{1}{2},\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{138}	$(\omega^2,-1,\omega^2)$	$\left(-\frac{1}{2},-1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,-\frac{\sqrt{3}}{2}\right)$
b_{139}	$(\omega^2,-\omega^2,1)$	$\left(-\frac{1}{2},\frac{1}{2},1,\ -\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2},0\right)$
b_{231}	$(-1,1,1)$	$(-1,1,1,\ 0,0,0)$
b_{232}	$(-1,\omega,\omega^2)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ 0,\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{233}	$(-1,\omega^2,\omega)$	$\left(-1,-\frac{1}{2},-\frac{1}{2},\ 0,-\frac{\sqrt{3}}{2},\frac{\sqrt{3}}{2}\right)$
b_{234}	$(-\omega,\omega^2,\omega^2)$	$\left(\frac{1}{2},-\frac{1}{2},-\frac{1}{2},\ -\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2},-\frac{\sqrt{3}}{2}\right)$
b_{235}	$(-\omega,1,\omega)$	$\left(\frac{1}{2},1,-\frac{1}{2},\ -\frac{\sqrt{3}}{2},0,\frac{\sqrt{3}}{2}\right)$

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
b_{236}	$(-\omega, \omega, 1)$	$\left(\frac{1}{2}, -\frac{1}{2}, 1, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0\right)$
b_{237}	$(-\omega^2, \omega, \omega)$	$\left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{238}	$(-\omega^2, 1, \omega^2)$	$\left(\frac{1}{2}, 1, -\frac{1}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}\right)$
b_{239}	$(-\omega^2, \omega^2, 1)$	$\left(\frac{1}{2}, -\frac{1}{2}, 1, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0\right)$
e_{11}	$(1, 2, 1)$	$(1, 2, 1, 0, 0, 0)$
e_{12}	$(\omega^2, 2, \omega)$	$\left(-\frac{1}{2}, 2, -\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0, \frac{\sqrt{3}}{2}\right)$
e_{13}	$(\omega, 2, \omega^2)$	$\left(-\frac{1}{2}, 2, -\frac{1}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}\right)$
e_{14}	$(\omega, 2\omega^2, \omega^2)$	$\left(-\frac{1}{2}, -1, -\frac{1}{2}, \frac{\sqrt{3}}{2}, -\sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
e_{15}	$(1, 2\omega^2, 1)$	$(1, -1, 1, 0, -\sqrt{3}, 0)$
e_{16}	$(\omega^2, 2\omega^2, \omega)$	$\left(-\frac{1}{2}, -1, -\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}, \frac{\sqrt{3}}{2}\right)$
e_{17}	$(\omega^2, 2\omega, \omega)$	$\left(-\frac{1}{2}, -1, -\frac{1}{2}, -\frac{\sqrt{3}}{2}, \sqrt{3}, \frac{\sqrt{3}}{2}\right)$
e_{18}	$(1, 2\omega, 1)$	$(1, -1, 1, 0, \sqrt{3}, 0)$
e_{19}	$(\omega, 2\omega, \omega^2)$	$\left(-\frac{1}{2}, -1, -\frac{1}{2}, \frac{\sqrt{3}}{2}, \sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
e_{21}	$(1, 1, 2)$	$(1, 1, 2, 0, 0, 0)$
e_{22}	$(\omega, \omega^2, 2)$	$\left(-\frac{1}{2}, -\frac{1}{2}, 2, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, 0\right)$
e_{23}	$(\omega^2, \omega, 2)$	$\left(-\frac{1}{2}, -\frac{1}{2}, 2, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0\right)$
e_{24}	$(\omega, \omega^2, 2\omega^2)$	$\left(-\frac{1}{2}, -\frac{1}{2}, -1, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
e_{25}	$(\omega^2, \omega, 2\omega^2)$	$\left(-\frac{1}{2}, -\frac{1}{2}, -1, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
e_{26}	$(1, 1, 2\omega^2)$	$(1, 1, -1, 0, 0, -\sqrt{3})$
e_{27}	$(\omega^2, \omega, 2\omega)$	$\left(-\frac{1}{2}, -\frac{1}{2}, -1, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, \sqrt{3}\right)$
e_{28}	$(\omega, \omega^2, 2\omega)$	$\left(-\frac{1}{2}, -\frac{1}{2}, -1, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}, \sqrt{3}\right)$
e_{29}	$(1, 1, 2\omega)$	$(1, 1, -1, 0, 0, \sqrt{3})$
e_{31}	$(2, 1, 1)$	$(2, 1, 1, 0, 0, 0)$
e_{32}	$(2, \omega, \omega^2)$	$\left(2, -\frac{1}{2}, -\frac{1}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
e_{33}	$(2, \omega^2, \omega)$	$\left(2, -\frac{1}{2}, -\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
e_{34}	$(2\omega^2, 1, 1)$	$(-1, 1, 1, -\sqrt{3}, 0, 0)$
e_{35}	$(2\omega^2, \omega, \omega^2)$	$\left(-1, -\frac{1}{2}, -\frac{1}{2}, -\sqrt{3}, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
e_{36}	$(2\omega^2, \omega^2, \omega)$	$\left(-1, -\frac{1}{2}, -\frac{1}{2}, -\sqrt{3}, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
e_{37}	$(2\omega, 1, 1)$	$(-1, 1, 1, \sqrt{3}, 0, 0)$
e_{38}	$(2\omega, \omega^2, \omega)$	$\left(-1, -\frac{1}{2}, -\frac{1}{2}, \sqrt{3}, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
e_{39}	$(2\omega, \omega, \omega^2)$	$\left(-1, -\frac{1}{2}, -\frac{1}{2}, \sqrt{3}, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1121}	$(2, -1, 1)$	$(2, -1, 1, 0, 0, 0)$
b_{1122}	$(2, -\omega, \omega^2)$	$\left(2, \frac{1}{2}, -\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1123}	$(2, -\omega^2, \omega)$	$\left(2, \frac{1}{2}, -\frac{1}{2}, 0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1124}	$(2, -\omega, \omega)$	$\left(2, \frac{1}{2}, -\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1125}	$(2, -\omega^2, 1)$	$\left(2, \frac{1}{2}, 1, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{1126}	$(2, -1, \omega^2)$	$\left(2, -1, -\frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{1127}	$(2, -\omega^2, \omega^2)$	$\left(2, \frac{1}{2}, -\frac{1}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1128}	$(2, -\omega, 1)$	$\left(2, \frac{1}{2}, 1, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{1129}	$(2, -1, \omega)$	$\left(2, -1, -\frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{2121}	$(-1, 2, 1)$	$(-1, 2, 1, 0, 0, 0)$
b_{2122}	$(-1, 2\omega, \omega^2)$	$\left(-1, -1, -\frac{1}{2}, 0, \sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2123}	$(-1, 2\omega^2, \omega)$	$\left(-1, -1, -\frac{1}{2}, 0, -\sqrt{3}, \frac{\sqrt{3}}{2}\right)$
b_{2124}	$(-1, 2\omega, \omega)$	$\left(-1, -1, -\frac{1}{2}, 0, \sqrt{3}, \frac{\sqrt{3}}{2}\right)$

Label	$\mathbb{C}^3$ vector	$\mathbb{R}^6$ vector
b_{2125}	$(-1, 2\omega^2, 1)$	$(-1, -1, 1, 0, -\sqrt{3}, 0)$
b_{2126}	$(-1, 2, \omega^2)$	$\left(-1, 2, -\frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{2127}	$(-1, 2\omega^2, \omega^2)$	$\left(-1, -1, -\frac{1}{2}, 0, -\sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2128}	$(-1, 2\omega, 1)$	$(-1, -1, 1, 0, \sqrt{3}, 0)$
b_{2129}	$(-1, 2, \omega)$	$\left(-1, 2, -\frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{1131}	$(2, 1, -1)$	$(2, 1, -1, 0, 0, 0)$
b_{1132}	$(2, \omega, -\omega^2)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1133}	$(2, \omega^2, -\omega)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1134}	$(2, \omega, -\omega)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\right)$
b_{1135}	$(2, \omega^2, -1)$	$\left(2, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{1136}	$(2, 1, -\omega^2)$	$\left(2, 1, \frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{1137}	$(2, \omega^2, -\omega^2)$	$\left(2, -\frac{1}{2}, \frac{1}{2}, 0, -\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right)$
b_{1138}	$(2, \omega, -1)$	$\left(2, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{1139}	$(2, 1, -\omega)$	$\left(2, 1, \frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{3131}	$(-1, 1, 2)$	$(-1, 1, 2, 0, 0, 0)$
b_{3132}	$(-1, \omega, 2\omega^2)$	$\left(-1, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3133}	$(-1, \omega^2, 2\omega)$	$\left(-1, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3134}	$(-1, \omega, 2\omega)$	$\left(-1, -\frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3135}	$(-1, \omega^2, 2)$	$\left(-1, -\frac{1}{2}, 2, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{3136}	$(-1, 1, 2\omega^2)$	$(-1, 1, -1, 0, 0, -\sqrt{3})$
b_{3137}	$(-1, \omega^2, 2\omega^2)$	$\left(-1, -\frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3138}	$(-1, \omega, 2)$	$\left(-1, -\frac{1}{2}, 2, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{3139}	$(-1, 1, 2\omega)$	$(-1, 1, -1, 0, 0, \sqrt{3})$
b_{2231}	$(1, 2, -1)$	$(1, 2, -1, 0, 0, 0)$
b_{2232}	$(1, 2\omega, -\omega^2)$	$\left(1, -1, \frac{1}{2}, 0, \sqrt{3}, \frac{\sqrt{3}}{2}\right)$
b_{2233}	$(1, 2\omega^2, -\omega)$	$\left(1, -1, \frac{1}{2}, 0, -\sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2234}	$(1, 2\omega, -\omega)$	$\left(1, -1, \frac{1}{2}, 0, \sqrt{3}, -\frac{\sqrt{3}}{2}\right)$
b_{2235}	$(1, 2\omega^2, -1)$	$(1, -1, -1, 0, -\sqrt{3}, 0)$
b_{2236}	$(1, 2, -\omega^2)$	$\left(1, 2, \frac{1}{2}, 0, 0, \frac{\sqrt{3}}{2}\right)$
b_{2237}	$(1, 2\omega^2, -\omega^2)$	$\left(1, -1, \frac{1}{2}, 0, -\sqrt{3}, \frac{\sqrt{3}}{2}\right)$
b_{2238}	$(1, 2\omega, -1)$	$(1, -1, -1, 0, \sqrt{3}, 0)$
b_{2239}	$(1, 2, -\omega)$	$\left(1, 2, \frac{1}{2}, 0, 0, -\frac{\sqrt{3}}{2}\right)$
b_{3231}	$(1, -1, 2)$	$(1, -1, 2, 0, 0, 0)$
b_{3232}	$(1, -\omega, 2\omega^2)$	$\left(1, \frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3233}	$(1, -\omega^2, 2\omega)$	$\left(1, \frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3234}	$(1, -\omega, 2\omega)$	$\left(1, \frac{1}{2}, -1, 0, -\frac{\sqrt{3}}{2}, \sqrt{3}\right)$
b_{3235}	$(1, -\omega^2, 2)$	$\left(1, \frac{1}{2}, 2, 0, \frac{\sqrt{3}}{2}, 0\right)$
b_{3236}	$(1, -1, 2\omega^2)$	$(1, -1, -1, 0, 0, -\sqrt{3})$
b_{3237}	$(1, -\omega^2, 2\omega^2)$	$\left(1, \frac{1}{2}, -1, 0, \frac{\sqrt{3}}{2}, -\sqrt{3}\right)$
b_{3238}	$(1, -\omega, 2)$	$\left(1, \frac{1}{2}, 2, 0, -\frac{\sqrt{3}}{2}, 0\right)$
b_{3239}	$(1, -1, 2\omega)$	$(1, -1, -1, 0, 0, \sqrt{3})$

IV. DISCUSSION

We have described a simple phase-adjusted realification procedure that embeds any finite set of rays in $\mathbb{C}^3$ into $\mathbb{R}^6$ in a way that preserves and reflects complex orthogonality. The construction is purely algebraic and uses only the phase freedom inherent in projective rays. Applied to the 165-ray configuration built from mutually unbiased bases, it yields a faithful orthogonal representation in $\mathbb{R}^6$ (after suitable phase choices), and, by trivial extension, in $\mathbb{R}^9$.

Table I lists all 165 rays explicitly in $\mathbb{C}^3$ together with their canonical realifications in $\mathbb{R}^6$. For applications requiring strict faithfulness (no extra orthogonality among non-orthogonal rays), one additionally picks phases θ_k as in Sec. II before applying Φ_0, which simply rotates each pair $(\Re z_j, \Im z_j)$ in its two-dimensional real subspace.

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